# CLT & LLN & Distributions recap

Com Eng Math II & Lab

# “Fundamental theorem” of statistics

- Law of Large Number (LLN)
- Central Limit Theorem (CLT)

# Convergence

- Converge in value
- Converge in probability
- Converge in distribution

- Converge in value

- Converge in probability

A sequence of random var $X_1 \ldots X_n$

With high prob, $X_n$ will be within $\varepsilon$ of $X$

- Converge in distribution

CDF $F(x)$, RV $x_1, \ldots$

# Chebychev Inequality

**Theorem 8.1 (Chebyshev Inequality)** Let $X$ be a discrete random variable with expected value $\mu = E(X)$, and let $\epsilon > 0$ be any positive real number. Then

$$P(|X - \mu| \geq \epsilon) \leq \frac{V(X)}{\epsilon^2} .$$

# Law of large numbers

- The average of many independent samples is (with high probability, converge in probability) close to the mean of the underlying distribution

$X_1, \ldots X_n$ iid, $X_i$ has mean $\mu$ and SD $\sigma$

Let $\bar{X}_n$ be the average of $X_1 \ldots X_n$

$$\bar{X}_n = \frac{X_1 + \ldots + X_n}{n} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

Note $\bar{X}_n$ is a RV (sum of RVs)

# Ex LLN

- Flip a fair coin. The observed rate of heads is $\bar{X}_n$

$$P(|\bar{X} - 0.5| < 0.1) \Rightarrow P(0.4 \le \bar{X}_n \le 0.6)$$

$n = 10 \quad P(0.4 \le \bar{X}_{10} \le 0.6) = 0.66$

$n = 50 \quad P(0.4 \le \bar{X}_{50} \le 0.6) = 0.88$

$n = 100 \quad P(0.4 \le \bar{X}_{100} \le 0.6 = 0.96$

$n = 500 \quad P(0.4 \le \bar{X}_{500} \le 0.6 = 0.9999941$

$n = 1000 \quad P(0.4 \le \bar{X}_{1000} \le 0.6 = 1$

# Conseqeunces of LLN

- Sampled mean is closed to the true mean at large n.
- Histograms are close to the actual pdf

- Notes
  - This LLN we talk about is the “weak” LLN
  - There is also ”strong” LLN which says
    - Converges “almost surely” to the mean

$$\text{Prob}\left(\lim_{n\to\infty} X_n = X\right) = 1$$

    - This is very similar to convergence in value. Hit a certain value and always stay there.

# Central Limit Theorem

$X_1 \ldots X_n$ iid $\mu, \sigma$

$$\bar{X}_n = \frac{X_1 + \ldots + X_n}{n}, \quad S_n = X_1 + \ldots + X_n$$

# Notes

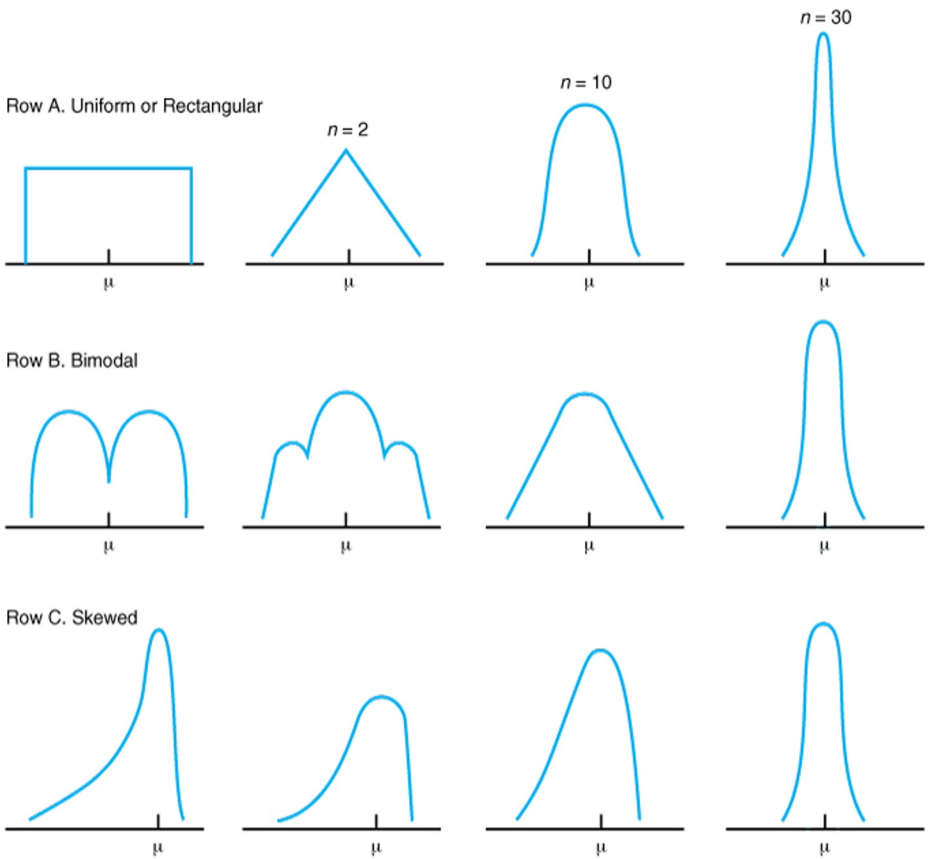
- $\bar{X}_n$ has same mean but smaller variance
- $\bar{X}_n$ and $S_n$ converges in distribution to the Normal dist

- "Large" some say 30… some say 6… Just eyeball

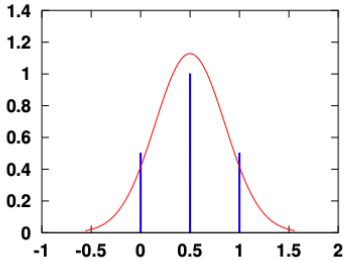
- This works on discrete distributions, even though it converges to Normal which is continuous. This is why convergence is defined on the CDF.
- There are other forms of CLT, that relaxes the independence or identically distributed assumption.

# Central Limit Theorem

- Any distribution works




http://www.people.vcu.edu/~albest/DENS580/DawsonTrapp/Chap4.htm

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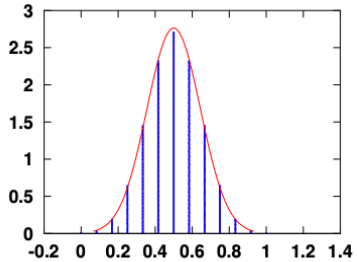


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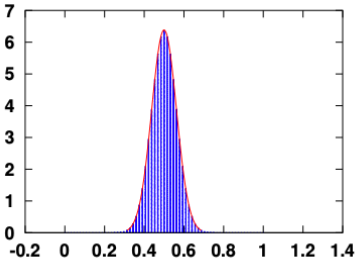


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https://ocw.mit.edu/courses/mathematics/18-05-introduction-to-probability-and-statistics-spring-2014/readings/MIT18_05S14_Reading6b.pdf

# Applications of CLT

- Flip a fair coin 100 times. Estimate the prob of more than 55 heads

# Confidence interval (polling)

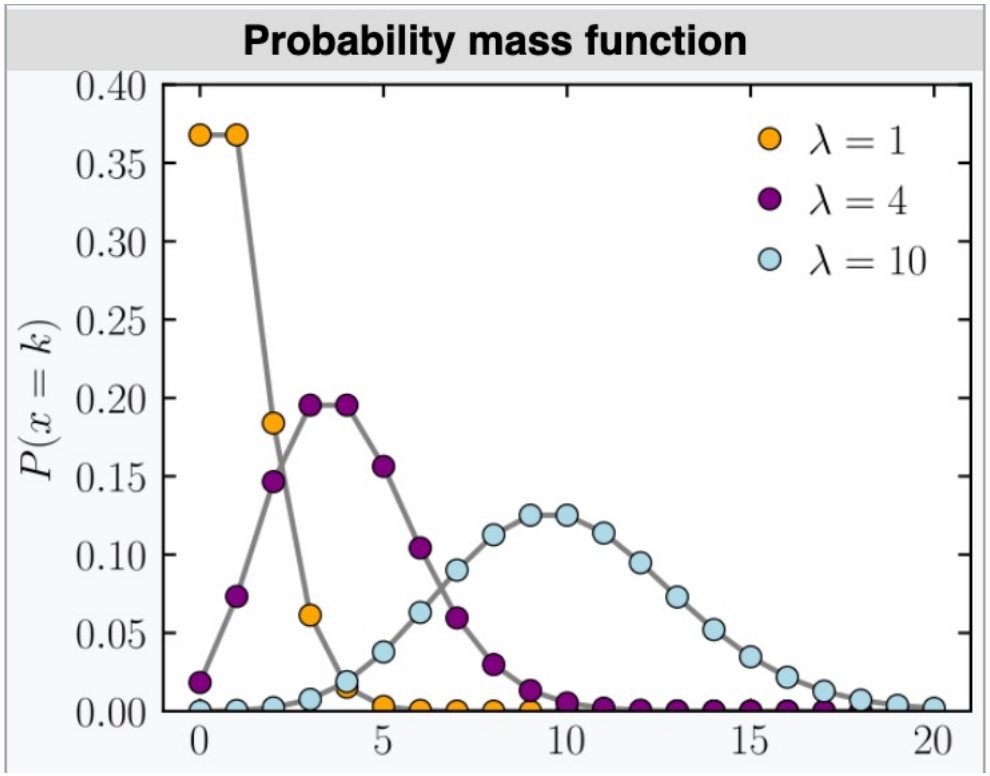
- Supposed there are choices A and B in a poll, and the population prefer A with prob $p_o$.
- You poll n people and compute the rate of A preference to be $\overline{X}$. Find the 95% confidence interval of candidate A preference.

$$\text{Prob}(\overline{X} \text{ is in } p_0 \pm e) = 0.95$$

# Poisson Distribution

- A discrete distribution that describes the number of occurance within a certain time.
- Parameterized by a rate parameter (Average number of occurance).

| | |
|---|---|
| **Parameters** | $\lambda \in (0, \infty)$ (rate) |
| **Support** | $k \in \mathbb{N}$ (Natural numbers starting from 0) |
| **PMF** | $\frac{\lambda^k e^{-\lambda}}{k!}$ |
| **CDF** | $\frac{\Gamma(\lfloor k+1 \rfloor, \lambda)}{\lfloor k \rfloor !}$, or $e^{-\lambda} \sum_{j=0}^{\lfloor k \rfloor} \frac{\lambda^j}{j!}$, or $Q(\lfloor k+1 \rfloor, \lambda)$ (for $k \geq 0$, where $\Gamma(x, y)$ is the upper incomplete gamma function, $\lfloor k \rfloor$ is the floor function, and $Q$ is the regularized gamma function) |
| **Mean** | $\lambda$ |
| **Variance** | $\lambda$ |



# Poisson distribution example

- CU pop bus comes with a rate of 3 per hour. Let's assume CU pop bus follows a Poisson distribution. What's the probability that there will be at least 1 bus in an hour?

# Common distributions (discrete)

- Benoulli distribution (binary)
- Binomial distribution (sum of independent benoulli trials)
- Geometric distribution (number of failures before first success)
  - Memoryless property
- Uniform
- Poisson (number of times an event happen in the interval)

# Common distributions (continuous)

- Uniform
- Exponential (wait time)
- Guassian/Normal (CLT)

# Relationships

- Benoulli distribution

- Geometric
- Exponential

- Binomial
- Poisson

# Summary

- LLN
- CLT
- Distributions